

6.3.12 Data Type

Computers store data in variables. Each variable will be of a pre-defined type. These types, or data types as they are more popularly known, include integers, floating point numbers, characters, strings, and arrays. Other data types include such things as date, timestamps, Boolean values, and varchar (variable character) formats. Some languages are more flexible and do not require that you explicitly assign a data type to each variable, and will assume the data type from the first value assigned to the variable. This has both advantages and disadvantages. Data types are also an integral part of database applications. Fields within a database should normally be assigned to a specific data type.

6.3.13 Debugging

The process of correcting the errors (or “bugs”) in a software program. Computer programs are long and complex pieces of code, and it is likely that there will be errors. A programmer has to go through the complete logic of the program and correct the errors to the maximum extent possible before releasing the software to end-users.

6.3.13 Debugger

A program that is used to trace errors in the code as programmers work on the program. It quickly identifies where exactly the errors (or bugs as they are called) are. The debugger will highlight the exact lines of code where problems are found. These bugs are usually related to errors in syntax representation in the programming language that’s being used. To find errors which are not obvious in the code and are errors more related to the logic of the algorithm used or related to interaction characteristics with the system, debuggers also support a step-by-step walk through of the program, which enable the programmer to see how the application reacts to each line of code and where the code breaks or the system crashes.

6.3.14 Floating Point

This is a data type that contains floating decimal points. Thus 37.001 is a floating point number, as compared to integers, which